

Build a Simple Multi-Step Agent Using LangChain or Rasa

Objective

This activity helps you move from AI theory to practical system design by building a multi-step agent that retrieves, processes, and outputs data using external tools. You'll practice chaining actions, handling errors gracefully, and debugging agent behavior in real time. Are you building something that works once—or something you can trust in production? This is your step toward engineering resilient, task-driven AI agents.

Tools Used:
LangChain or Rasa

Instructions for learners

Use LangChain or Rasa to build a simple multi-step AI agent that performs a sequence of 2–3 tasks (e.g., retrieve data, analyze it, and output a summary). Your agent should use at least one external tool or function (such as a web search API or document parser).

To simulate production robustness, include a fallback or error-handling mechanism. For example, if the tool call fails, your agent should respond with a safe default message or retry logic.

In addition, add at least one debug print or logging step to show how the agent transitions between actions. This will help you trace issues when the output doesn't match your expectations.

Submission

You may use templates or starter code, but the action logic should be your own. Submit your annotated code with comments explaining each part.